Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 19: compare collected examples with intended users.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Identify selection and coverage bias. Try the recall before opening notes.

Your learning target

I can identify selection and coverage **bias**.

I can compare groups before combining results.

I can explain why more **bias**ed data may not help.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

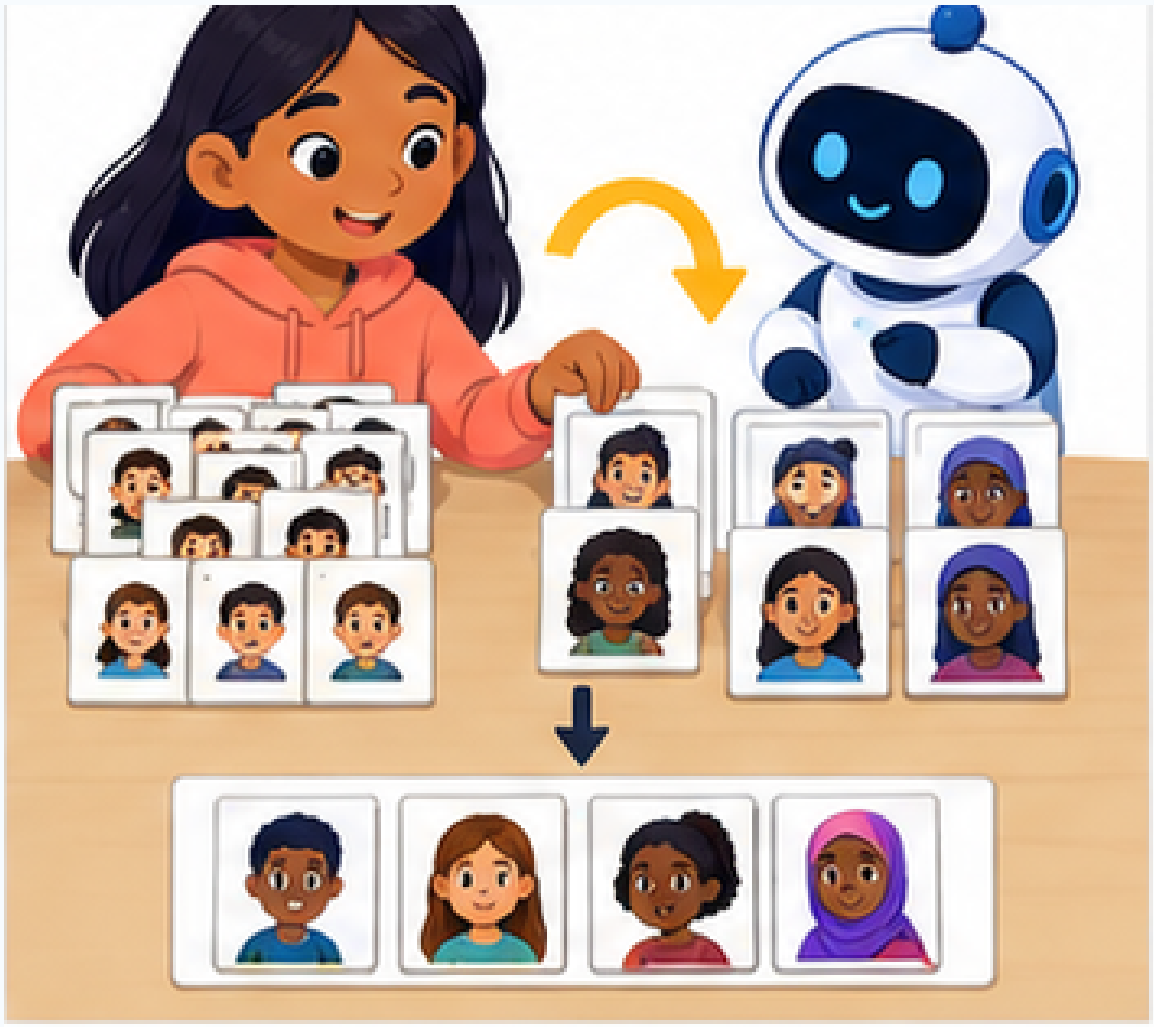
Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 20: Repair **Bias** in Data

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.



THINKER MISSION

Redesign an imbalanced speech dataset and specify **retest**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> What is the difference between a population and a sample?

This week's end goal

By the end, I can distinguish a **bias**ed collection **process** from a mere small dataset.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Repair Bias in Data case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use repair **bias** in data to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents. If only club members are surveyed, an 80 percent result misses a group with a different pattern.

Examine who is included before treating a larger sample as better.

STUDENT LESSON · 3 OF 18

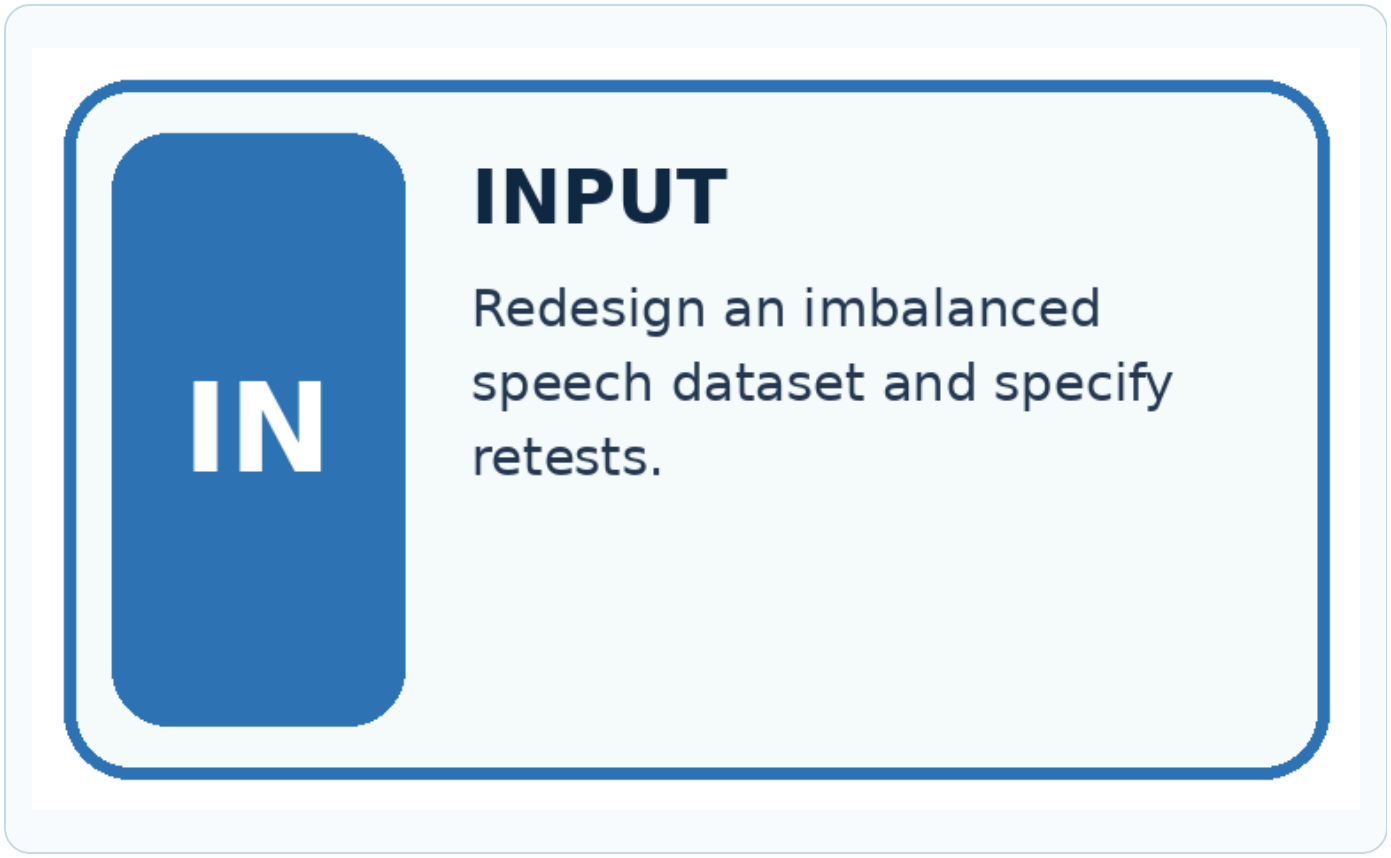
See how it works

> Follow the mechanism for Repair Bias in Data. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

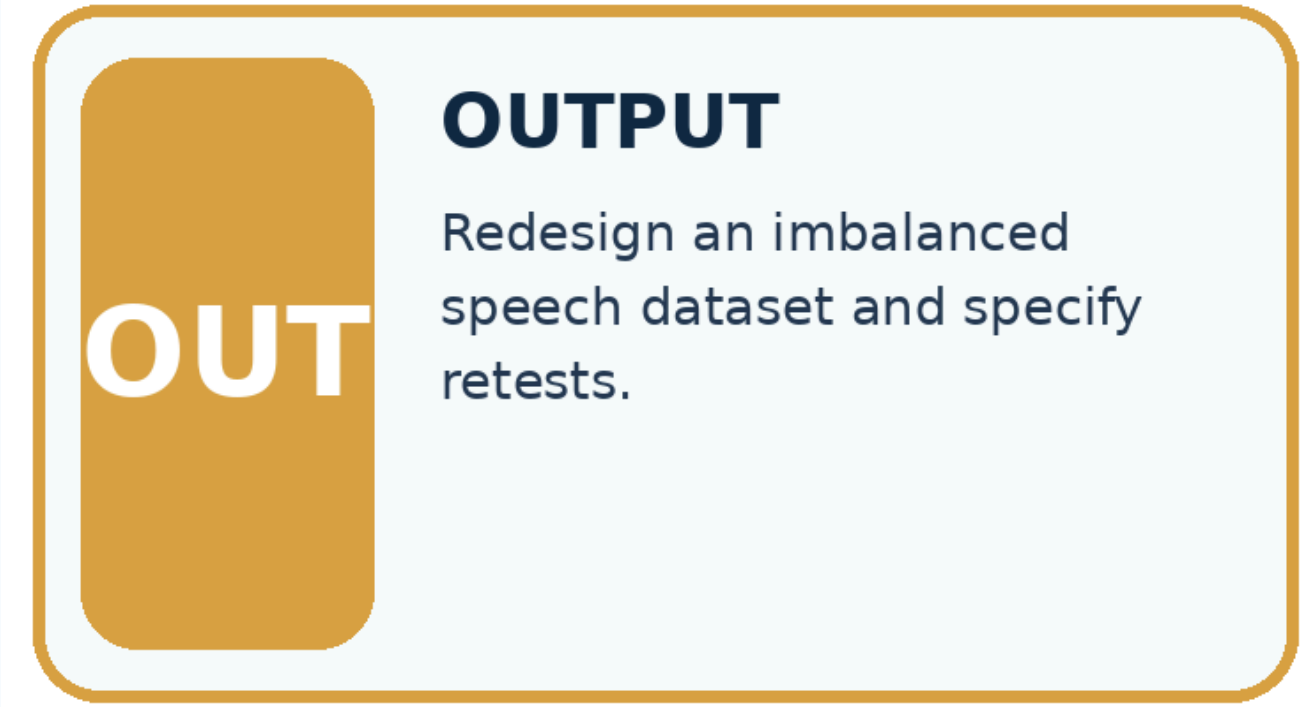
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Redesign an imbalanced speech dataset and specify retests.

↓


PROCESS
Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
club_yes, club_n = 8, 10
other_yes, other_n = 2, 10
print(club_yes / club_n)
print(other_yes / other_n)
print((club_yes + other_yes) / (club_n + other_n))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Club sample	8 of 10	80 percent
Other sample	2 of 10	20 percent
Combined sample	10 of 20	50 percent

Read each row from left to right and connect the evidence to the stated rule.

The combined observed sample has 50 percent Yes. This is not automatically a population estimate: we also need to know how the groups were selected and how many people each represents.

Do not average percentages without considering their denominators.

Combine counts, not just percentages

The club has 8 Yes out of 10, and the other group has 2 Yes out of 10. Combine the Yes counts to get 10 and combine the response counts to get 20; the combined observed rate is 10 divided by 20, or 50 percent.

When group sizes differ, simply averaging their percentages can give the wrong combined rate. A correct combined calculation still describes the collected sample, not automatically the whole school.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

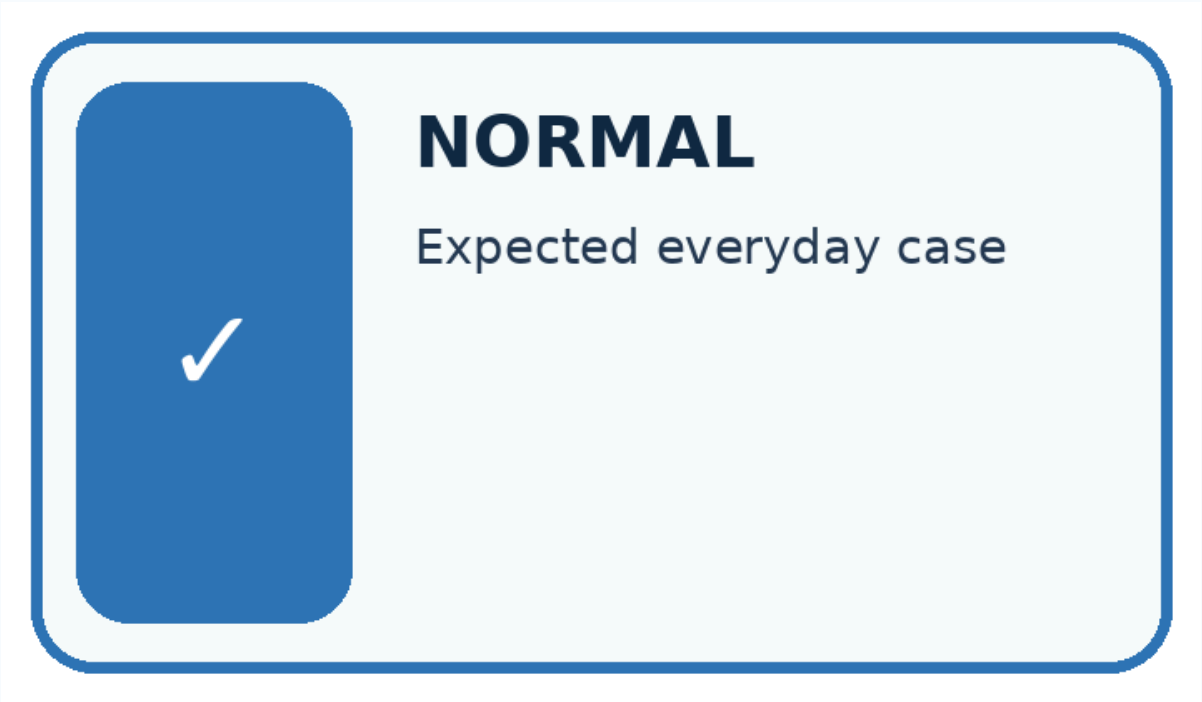
> Compare the supplied cases for Repair Bias in Data. Identify the detail that changes the conclusion.

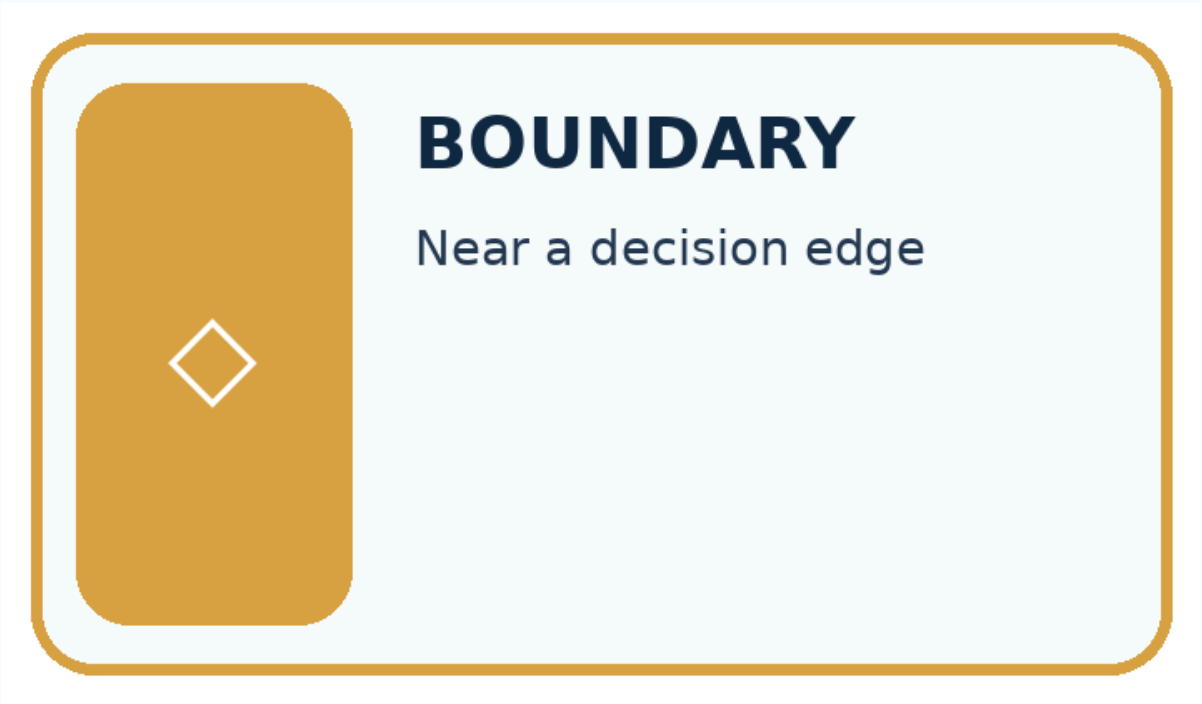
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

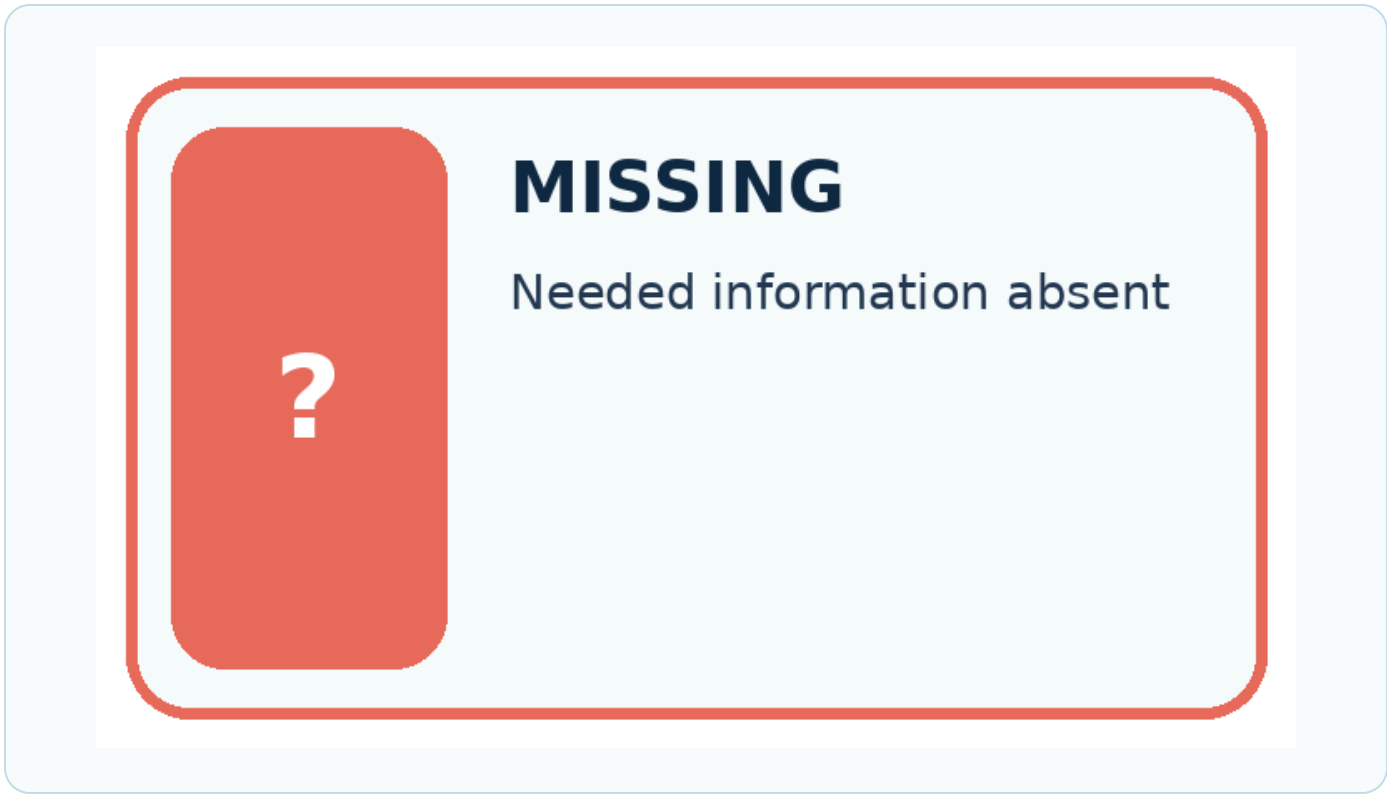
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Recruit more club volunteers only.

Predicted result

Prediction: Sample size increases without fixing missing representation of other learners.

Why the result follows

Reason: The added respondents come from the already overrepresented group.

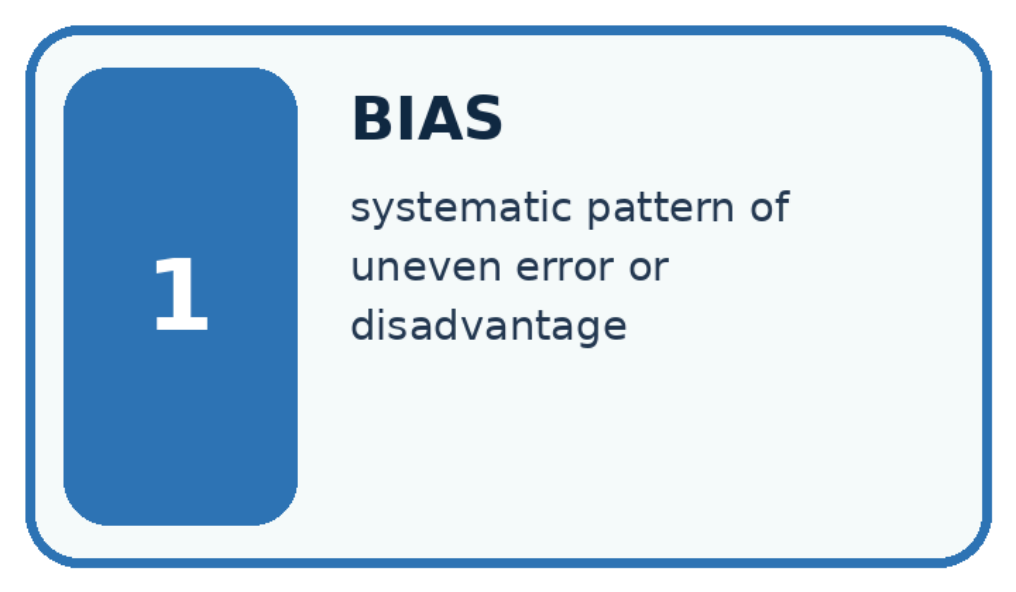
Words that unlock the idea

> Use the vocabulary of Repair Bias in Data to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

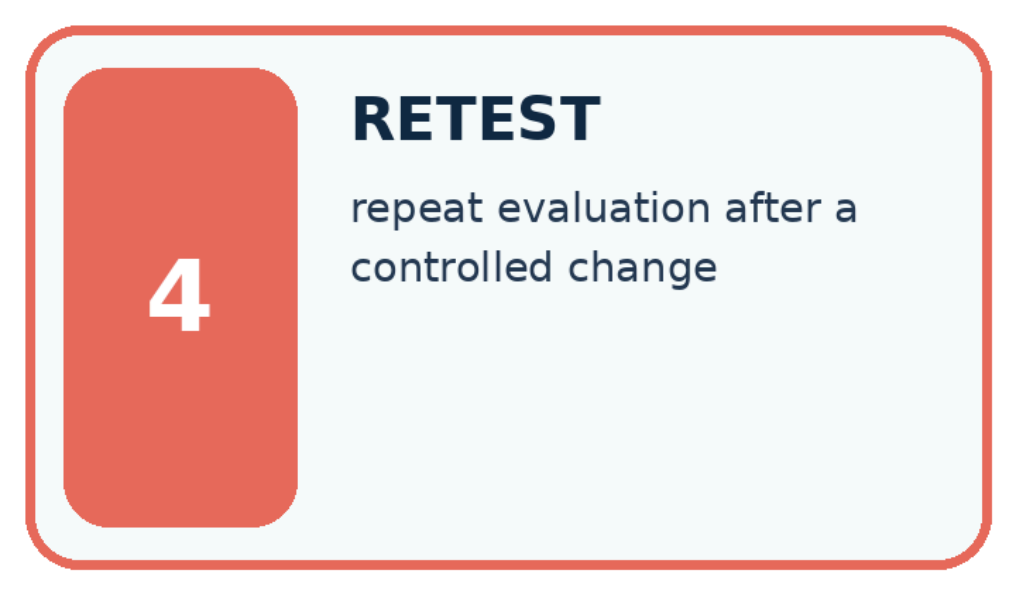
REBALANCE

change coverage across groups or conditions

3

SUBGROUP

meaningful part of the population



Word	My own example	Not this / boundary
bias	A systematic tendency in data, decisions or outcomes that can distort a result or disadvantage people.	
rebalance	Change the relative representation of groups or classes in data using a stated method.	
subgroup	A defined subset of cases considered separately in analysis.	
retest	Run a test again after a change or under specified new conditions.	

Words used in the closer explanation

Word	Meaning
Selection bias	Systematic distortion caused by who enters the data
Undercoverage	Leaving part of the target group out of selection
Nonresponse	A selected participant does not respond

STUDENT LESSON · 6 OF 18

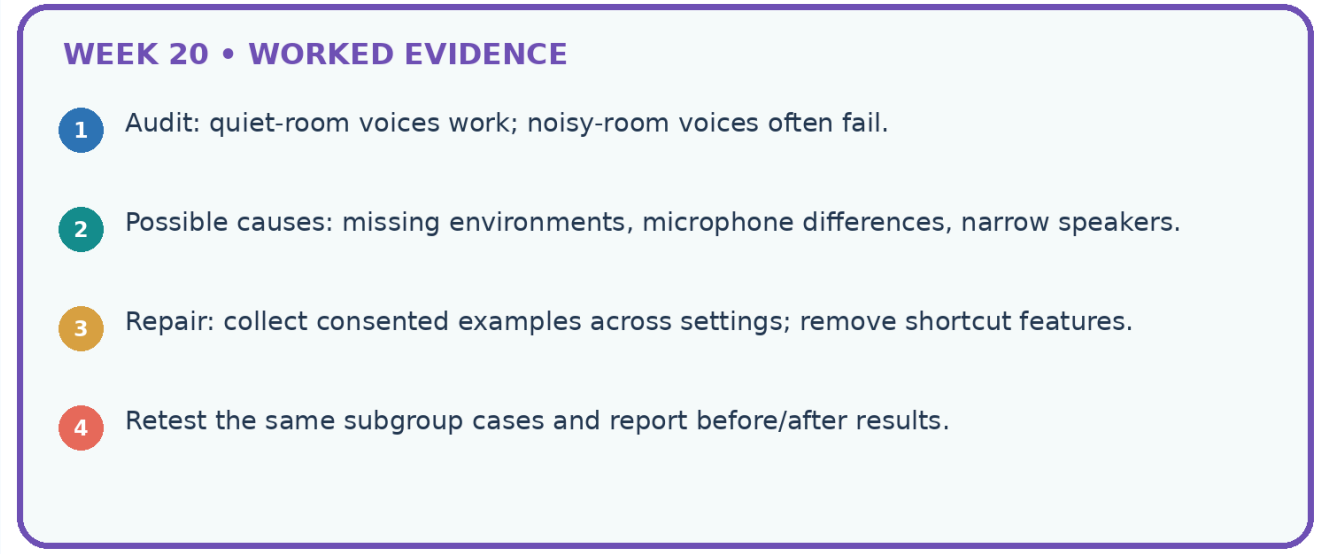
Follow a worked example

> Follow the worked trace for Repair Bias in Data, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Audit: quiet-room voices work; noisy-room voices often fail.

2. Possible causes: missing environments, microphone differences, narrow speakers.

3. Repair: collect consented examples across settings; remove shortcut features.

4. **Retest** the same **subgroup** cases and report before/after results.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Suppose 90 club respondents give 72 Yes and 10 others give 2 Yes. What is the combined observed proportion?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Repair Bias in Data

A fictional school polls learners about adding comic recommendations.

The facts we will use

> **Supplied facts**
>
> Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

> **Supplied facts**
>
> The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club:12/15=80%; others:3/15=20%.



2. Respondent total

15 Yes out of 30 respondents =50%.



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan including equality, documented invitations and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group rates	Club:12/15=80%; others:3/15=20%.
2. Respondent total	15 Yes out of 30 respondents =50%.
3. Check representation	Respondents split 50/50, but the grade is 25% club and 75% other.
4. Repair collection	Plan including equality, documented invitations and report nonresponse by group.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

What this example does not establish

Keep the stated limit

Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Keep the stated limit

Read the labelled picture: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Keep the stated limit

Change one thing: Recruit more club volunteers only. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Repair Bias in Data claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Adding more total data automatically repairs bias. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

> **Club members favor coding more often. Does club membership necessarily cause that preference?**

STUDENT LESSON · 8 OF 18

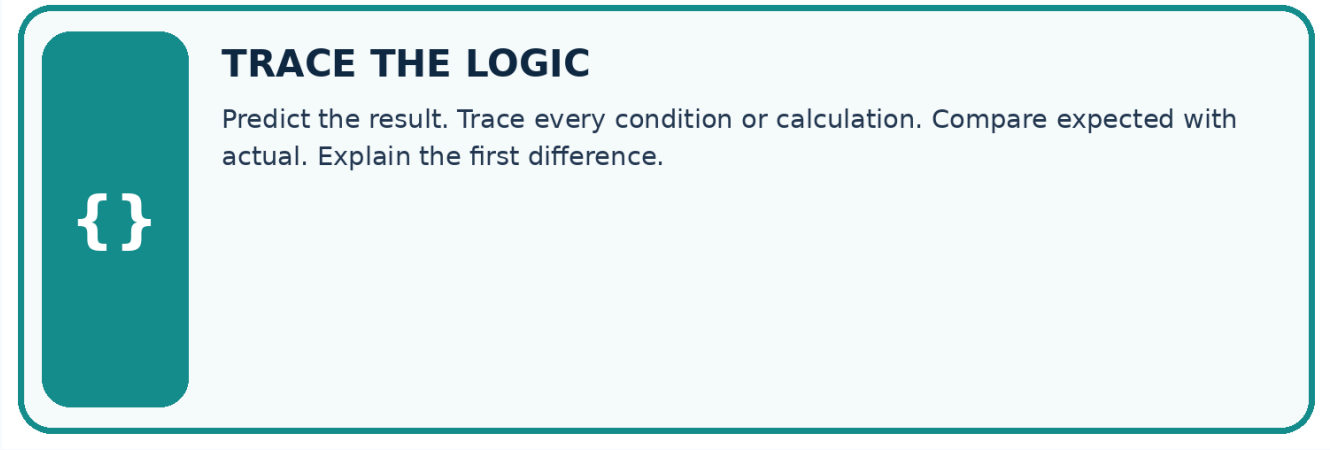
Practise together

> Practise Repair Bias in Data with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; **RETEST** EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Surveying another 100 people from the same undercovered source may make the sample larger without fixing the selection problem. Improve access to missing groups and record **nonresponse**; do not simply duplicate existing rows.

Practise with less help: Combine counts

Use workbook W2 for the connected example "**Selection bias** and coverage". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

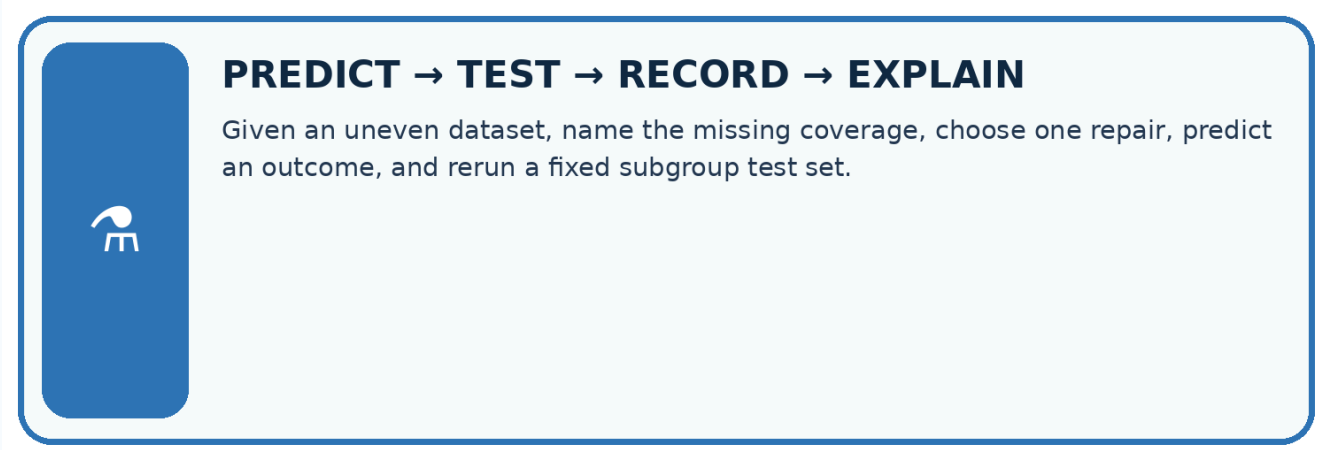
Run the investigation

> Investigate Repair Bias in Data. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Given an uneven dataset, name the missing coverage, choose one repair, predict an outcome, and rerun a fixed **subgroup** test set.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

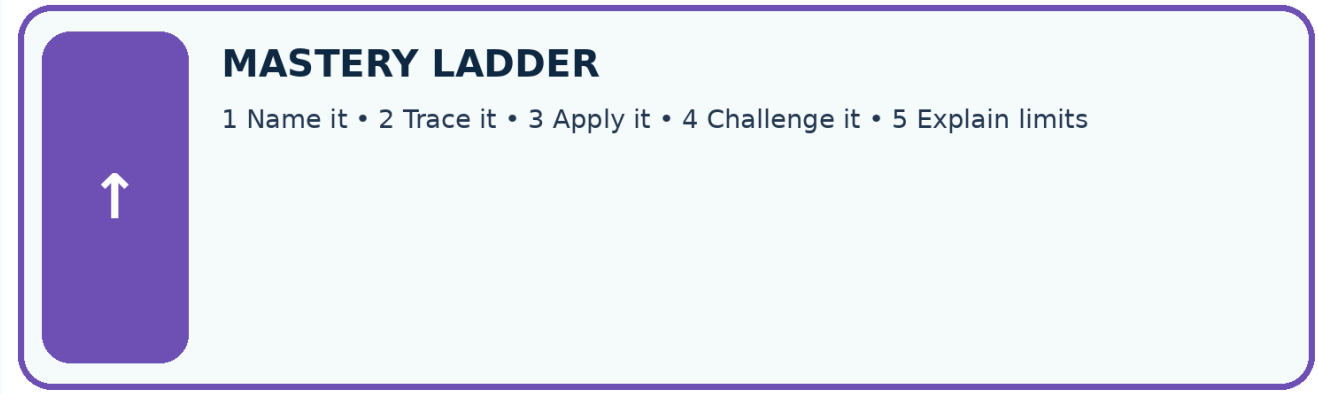
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Repair Bias in Data. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.

TRANSPARENT BOOK RECOMMENDER

Change feature coverage or weights only after locating which reader profiles are underserved.

THIS WEEK'S CHECKPOINT Change feature coverage or weights only after locating which reader profiles are underserved.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Redesign an imbalanced speech dataset and specify retests.

Try a more demanding case

CHALLENGE Redesign an imbalanced speech dataset and specify **retest**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Repair Bias in Data case for supported practice. Repeated examples do not become fresh assessments.

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

The question

Propose repair and evaluation.

Worked explanation

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Repair coverage with varied examples

Evidence or stage	Value or result
Adventure readers	18
Poetry readers	2

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Change feature coverage or weights only after locating which reader profiles are underserved.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Repair Bias in Data, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional school polls learners about adding comic recommendations.

By the end, I can	How I will show it
Identify selection and coverage bias	Explain the deciding step, show a trace or test, and state one limit.
Compare groups before combining results	Explain the deciding step, show a trace or test, and state one limit.
Explain why more biased data may not help	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Repair Bias in Data without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Identify selection and coverage **bias**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional school polls learners about adding comic recommendations.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Plan including equality, documented invitations and report **nonresponse** by group.

1. Group rates

2. Respondent total

3. Repair collection

4. Check representation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair collection Plan including equality, documented invitations and report **nonresponse** by group.

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Recruit more club volunteers only. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Sample size increases without fixing missing representation of other learners.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Repair a missing building block

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

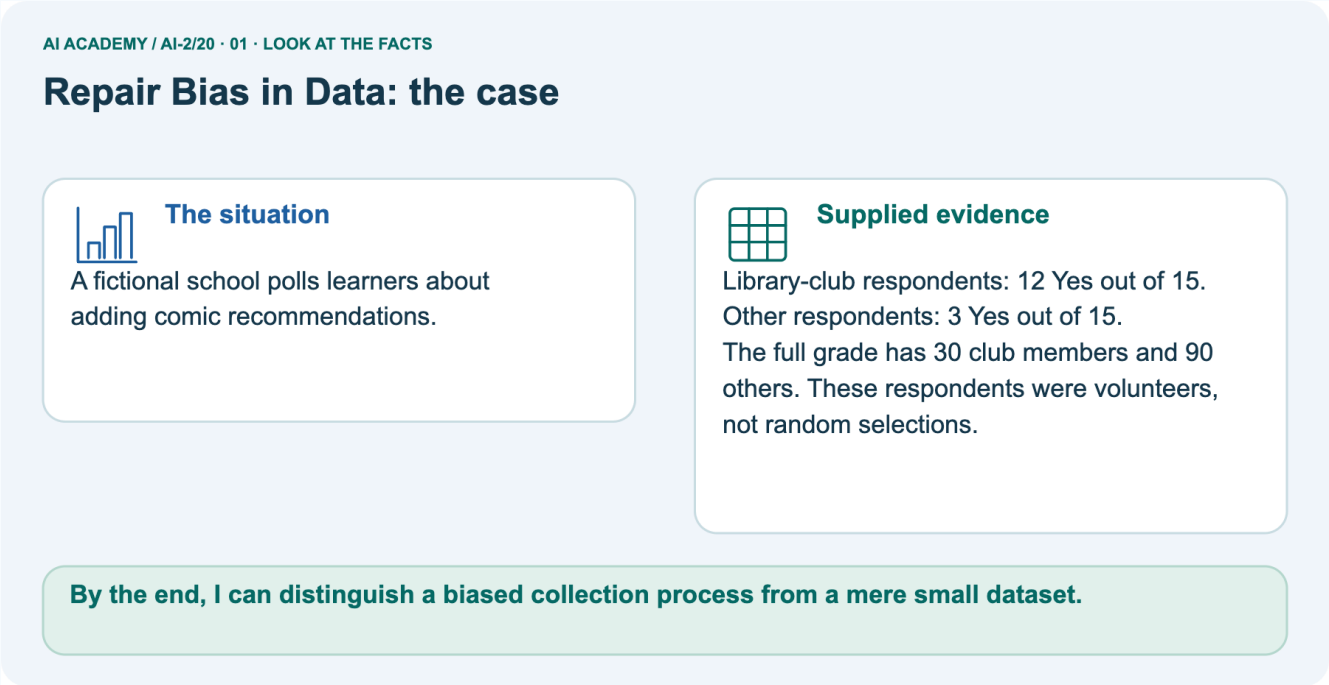
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Repair Bias in Data graphic using its labels, deciding evidence and relevant limitation.

01. Repair Bias in Data: the case



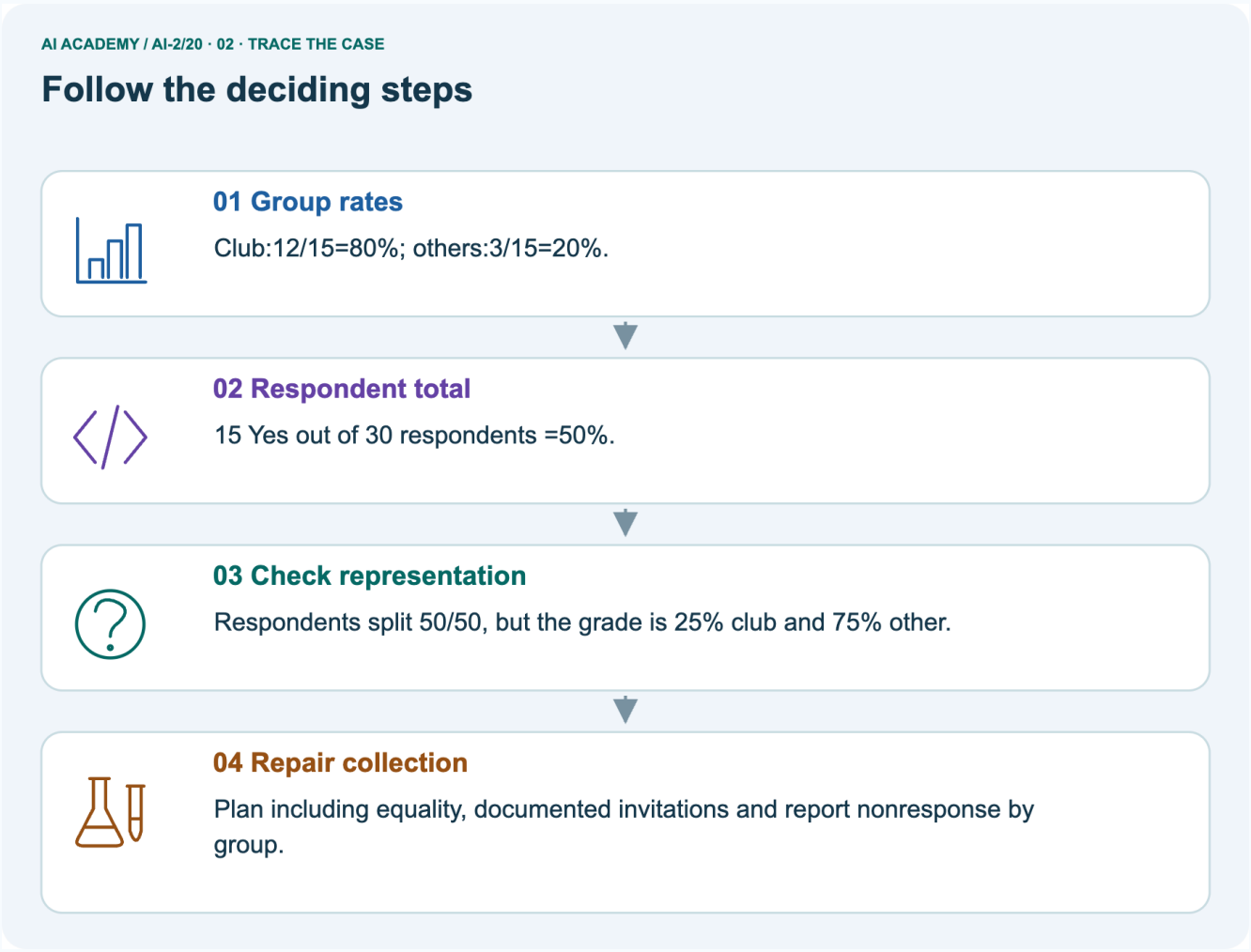
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

These respondents were volunteers, not random selections.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

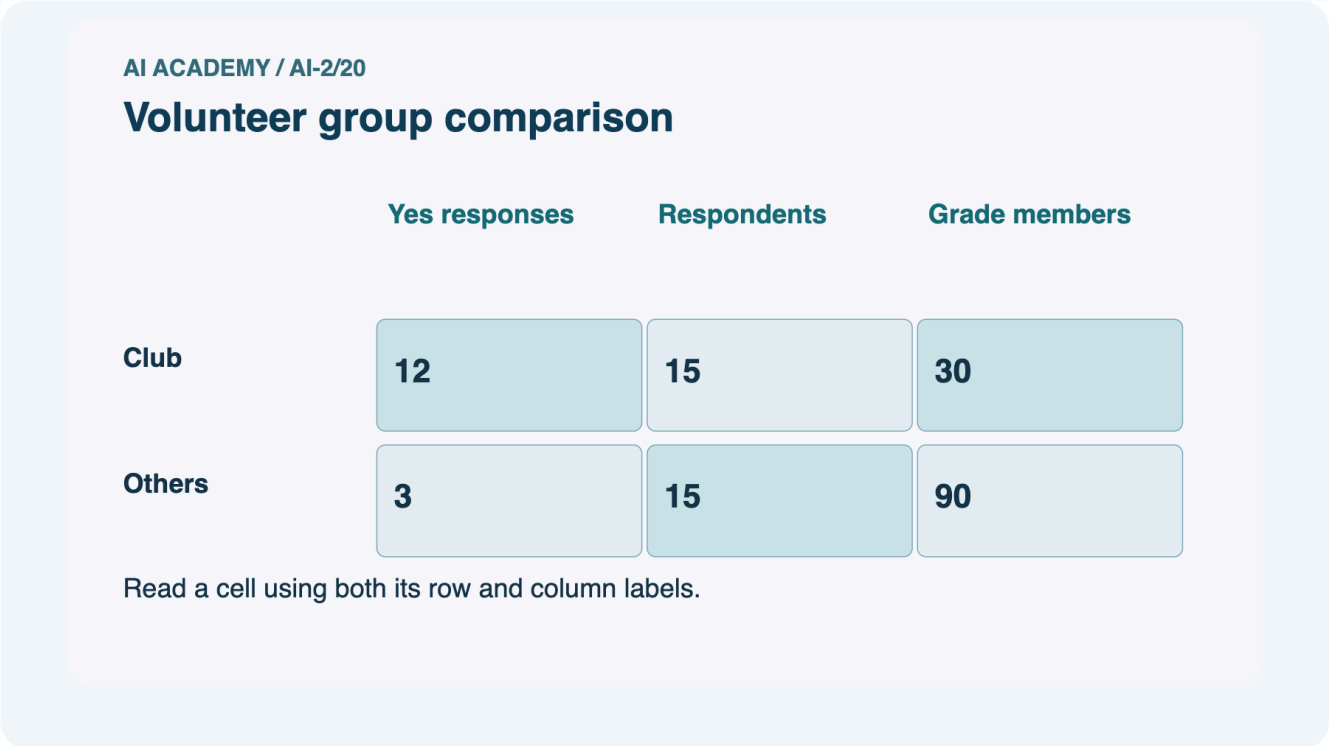


Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan including equality, documented invitations and report **nonresponse** by group.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Volunteer group comparison

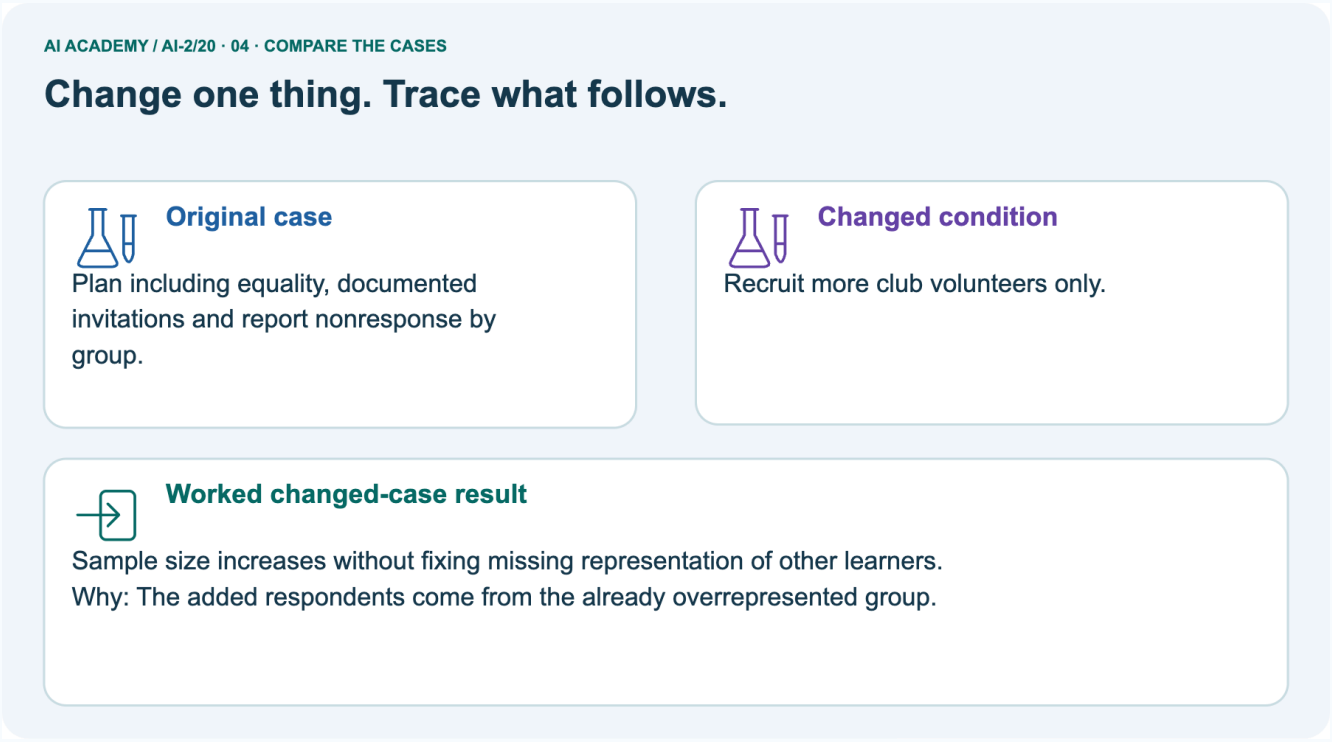


Follow the labelled picture with the matching explanation. Open the vector diagram

Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



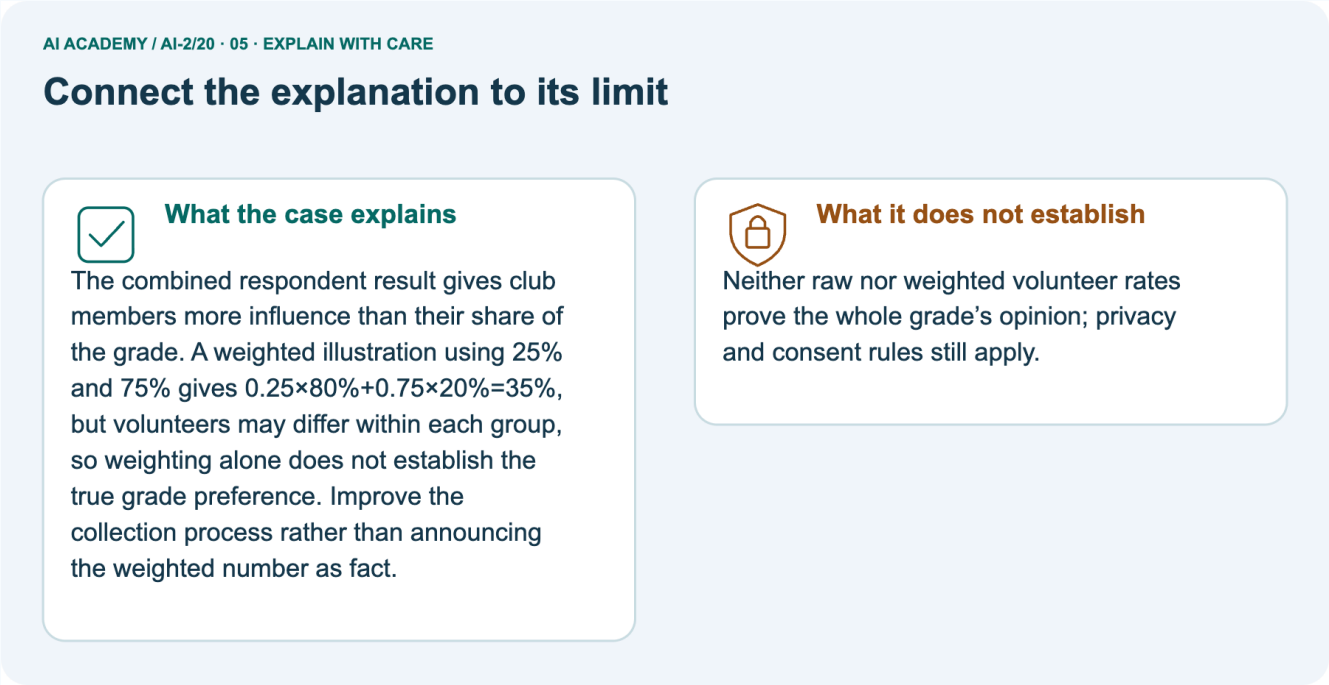
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Plan including equality, documented invitations and report **nonresponse** by group.
Changed condition: Recruit more club volunteers only.

Expected result: Sample size increases without fixing missing representation of other learners.
Reason: The added respondents come from the already overrepresented group.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

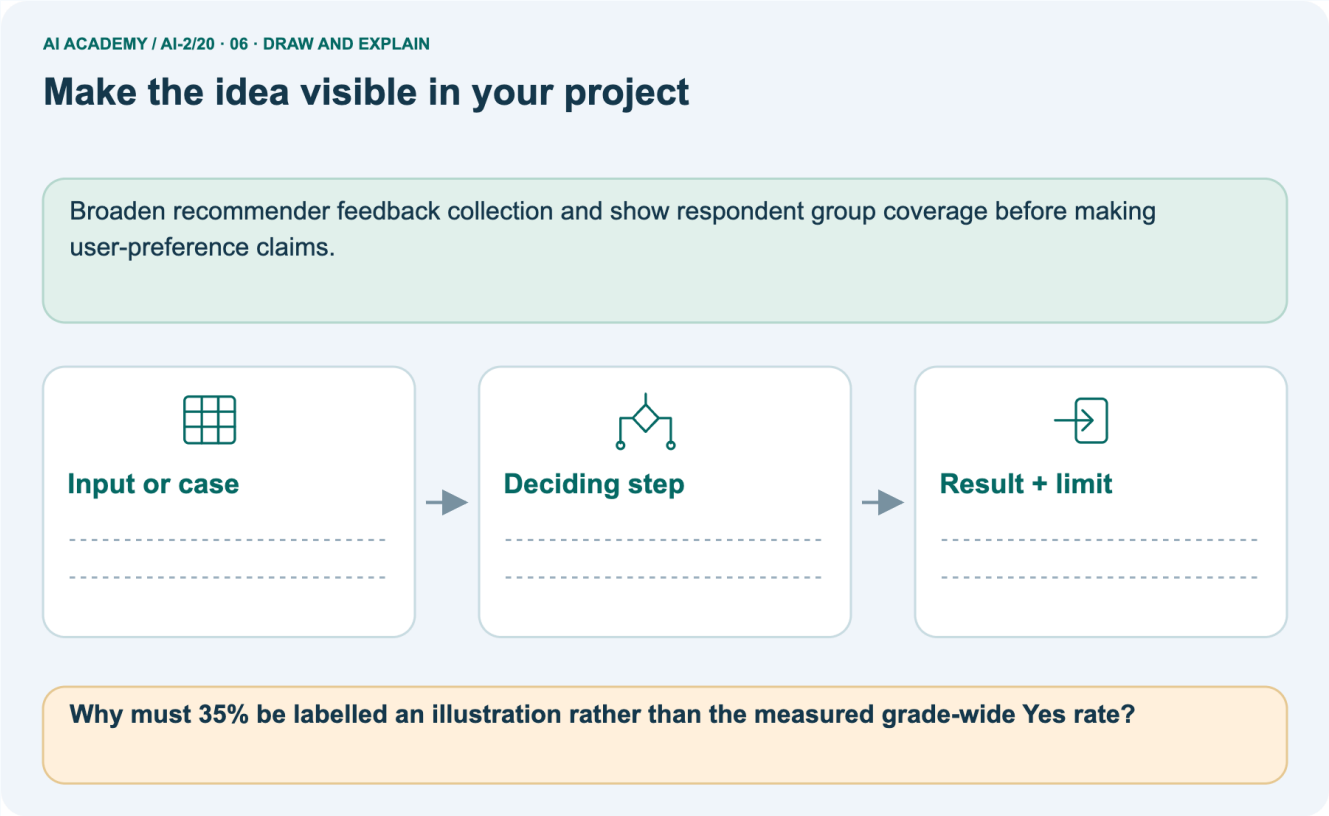
The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.